

Solution:

(a) Assume $\beta_i = (\alpha_1 + \dots + \alpha_i)\mu$, $\beta_0 = 0$,

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty \Pr[X \geq t] dt \\ &= \sum_{i=1}^k \int_{\beta_{i-1}}^{\beta_i} \Pr[X \geq t] dt + \int_{\beta_k}^\infty \Pr[X \geq t] dt\end{aligned}$$

Since $\Pr[X \geq t]$ is non-increase in t ,

$$\begin{aligned}\int_{\beta_{i-1}}^{\beta_i} \Pr[X \geq t] dt &\leq \Pr[X \geq \beta_{i-1}](\beta_i - \beta_{i-1}) \\ &= p_i \alpha_i \mu\end{aligned}$$

Thus,

$$\begin{aligned}\mathbb{E}[X] &\leq \sum_{i=1}^k p_i \alpha_i \mu \\ \sum_{i=1}^k \alpha_i p_i &\leq 1\end{aligned}$$

(b) The probability that the Monte Carlo algorithm fails is $p_1 p_2 p_3$, from (a),

$$\alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3 \leq 1 \quad (\alpha_1 p_1, \alpha_2 p_2, \alpha_3 p_3 \geq 0)$$

Thus,

$$\begin{aligned}\alpha_1 p_1 \alpha_2 p_2 \alpha_3 p_3 &\leq \frac{1}{27} \\ \Pr(\text{error}) = p_1 p_2 p_3 &\leq \frac{1}{27 \alpha_1 \alpha_2 \alpha_3}\end{aligned}$$

(c) Let $\alpha_1 = \frac{c}{3}, \alpha_2 = \frac{c}{2}, \alpha_3 = \frac{c}{1}$, the Monte Carlo algorithm repeats for $\lfloor \frac{t}{3c} \rfloor$ rounds, each round takes $3c\mu$ steps. The total number of steps is bounded by,

$$\left\lfloor \frac{t}{3c} \right\rfloor \cdot 3c\mu \leq t\mu$$

The probability that the algorithm fails is,

$$\Pr(\text{error}) \leq \left(\frac{6}{27c^3}\right)^{\lfloor \frac{t}{3c} \rfloor} = O\left(\left(\frac{6}{27c^3}\right)^{\frac{t}{3c}}\right)$$

Let $c = \frac{\sqrt[3]{6}e}{3} \approx 1.646$, we get,

$$\Pr(\text{error}) \leq O(0.545^t)$$

- (d) Let $\alpha_i = \frac{1}{k+1-i}$ for $i = 1, \dots, k$. The Monte Carlo algorithm repeats for $\lfloor \frac{t}{k} \rfloor$ rounds, each round takes $k\mu$ steps. The total number of steps is bounded by $t\mu$. The probability that each round fails is,

$$\prod_{i=1}^k p_i \leq k^{-k} \prod_{i=1}^k \alpha_i^{-1} = k^{-k} k!$$

The probability that the algorithm fails is,

$$\Pr(\text{error}) \leq (k^{-k} k!)^{\lfloor \frac{t}{k} \rfloor} = O((k^{-k} k!)^{\frac{t}{k}})$$

$\Pr(\text{error})$ is decreasing when $k > 0$, let $k \geq 32$,

$$\Pr(\text{error}) \leq O(0.3997^t) \leq O(0.4^t)$$